

3.7 Recommendations:

Bearing Capacity of Bored Pile for

Bore Hole No.	Dia. of pile (mm)	Length (M)	Skin friction (KN)	Skin friction (Ton)
1	600	21	708	72.171
	600	22.5	981	100.00
	600	24.0	1307	133.23
	500	21	590	60.143
	500	22.5	817	83.282
	500	24.0	1089	111.01
	450	21	532	54.23
	450	22.5	736	75.025
	450	24.0	981	100.00

Bore Hole No.	Dia. of pile (mm)	Length (M)	Skin friction (KN)	Skin friction (Ton)
2	600	21	752	76.656
	600	22.5	1025	104.49
	600	24.0	1369	139.55
	500	21.0	627	63.914
	500	22.5	854	87.054
	500	24.0	1141	116.31
	450	21.0	567	57.798
	450	22.5	771	78.593
	450	24.0	1029	104.89

Bore Hole No.	Dia. of pile (mm)	Length (M)	Skin friction (KN)	Skin friction (Ton)
3	600	21	463	47.197
	600	22.5	948	96.636
	600	24.0	1321	134.66
	500	21.0	537	54.74
	500	22.5	791	80.632
	500	24.0	1102	112.33
	450	21.0	486	49.541
	450	22.5	715	72.885
	450	24.0	995	101.43

Bore Hole No.	Dia. of pile (mm)	Length (M)	Skin friction (KN)	Skin friction (Ton)
4	600	21	819	83.486
	600	22.5	1124	114.58
	600	24.0	1539	156.88
	500	21	682	69.521
	500	22.5	936	95.413
	500	24.0	1282	130.68
	450	21	616	62.793
	450	22.5	845	86.137
	450	24.0	1156	117.84

Bore Hole No.	Dia. of pile (mm)	Length (M)	Skin friction (KN)	Skin friction (Ton)
5	600	21	782	79.715
	600	22.5	1087	110.81
	600	24.0	1548	157.8
	500	21	651	66.361
	500	22.5	905	92.253
	500	24.0	1289	131.4
	450	21	587	59.837
	450	22.5	816	83.18
	450	24.0	1162	118.45

- Regarding the deep foundation, the required values of the skin friction as well as the end bearing capacities of the pile (Bore type & Driven Pile) are provided in the Appendix: 2.

- The Carrying capacities of the pile, to be obtain/calculated from the appendix's values must be confirmed from carrying out the load test, covering the entire area.

- *[Note: The designer may select any other alternative type, depth as well as the bearing capacity of the foundation or the method of improving the topsoil according to his requirements, in the light of the information provided in this report.]*

CLOSURE:

We appreciate the opportunity to perform this investigation for you and have pleasure in submitting this report. Please contact us when we can be further service to you.

For MEGHLA SOIL & FOUNDATION

Report Prepared by

SPT Correction																			
Project :																	Borehole No		5
Client:																	Water Table		-0.00m
Location:			SITE:- PLOT NO. - 4356, ROAD NO. - 23, BLOCK - N, BASHUNDHARA RESIDENTIAL AREA, DHAKA, BANGLADESH.																
Depth in meter measured from EGL.			Field SPT	Soil Type	Unit Wt. of soil (kN/m ³)	Unit Wt. of water (kN/m ³)	Effective Unit Wt. of soil (kN/m ³)	Effective overburden pressure σ_v (kN/m ²)	Avg. Effective overburden pressure σ_v (kN/m ²)	C_u	Hammer efficiency, E_d	Borehole diameter correction, C_b	Sampler correction, C_s	Rod length correction, C_r	Corrected SPT (N_{60}) after Field procedure	Corrected SPT due to Overburden Pressure (N_{60})	Correction of SPT Value for water table (dilatancy) Value N_{60}	N_{60}	N_{60}
from	to	Average																	
0.0	1.5	0.8	2	Sand	20.0	9.81	10.19	15.29	7.64	1.862	0.6	1	1	0.75	2	3	3	2	3
1.5	3.0	2.3	2	Sand	20.0	9.81	10.19	30.57	22.93	1.494	0.6	1.00	1	0.75	2	2	2	2	2
3.0	4.5	3.8	2	Sand	20.0	9.81	10.19	45.86	38.21	1.323	0.6	1.00	1	0.85	2	2	2	2	2
4.5	6.0	5.3	2	Sand	20.0	9.81	10.19	61.14	53.50	1.211	0.6	1.00	1	0.95	2	2	2	2	3
6.0	7.5	6.8	3	Silt	17.0	9.81	7.19	53.93	48.53	1.244	0.6	1.00	1	0.95	3	4	4	3	4
7.5	9.0	8.3	1	Silt	17.0	9.81	7.19	64.71	59.32	1.176	0.6	1.00	1	0.95	1	1	1	1	1
9.0	10.5	9.8	1	Silt	17.0	9.81	7.19	75.50	70.10	1.121	0.6	1.00	1	1	1	1	1	1	1
10.5	12.0	11.3	2	Silt	17.0	9.81	7.19	86.28	80.89	1.073	0.6	1.00	1	1	2	2	2	2	2
12.0	13.5	12.8	4	Silt	17.0	9.81	7.19	97.07	91.67	1.031	0.6	1.00	1	1	4	4	4	4	4
13.5	15.0	14.3	4	Silt	17.0	9.81	7.19	107.85	102.46	0.994	0.6	1.00	1	1	4	4	4	3	4
15.0	16.5	15.8	5	Clay	18.0	9.81	8.19	135.14	128.99	0.917	0.6	1.00	1	1	5	5	5	4	5
16.5	18.0	17.3	7	Clay	18.0	9.81	8.19	147.42	141.28	0.886	0.6	1.00	1	1	7	6	6	5	7
18.0	19.5	18.8	40	Sand	20.0	9.81	10.19	198.71	191.06	0.785	0.6	1.00	1	1	40	31	23	27	34
19.5	21.0	20.3	46	Sand	20.0	9.81	10.19	213.99	206.35	0.760	0.6	1.00	1	1	46	35	25	30	38
21.0	22.5	21.8	94	Sand	20.0	9.81	10.19	229.28	221.63	0.736	0.6	1.00	1	1	94	69	42	59	75
22.5	24.0	23.3	155	Sand	20.0	9.81	10.19	244.56	236.92	0.713	0.6	1.00	1	1	155	111	63	95	121
24.0	25.5	24.8	165	Sand	20.0	9.81	10.19	259.85	252.20	0.692	0.6	1.00	1	1	165	114	65	98	125
25.5	27.0	26.3	175	Sand	20.0	9.81	10.19	275.13	267.49	0.673	0.6	1.00	1	1	175	118	66	101	128
27.0	28.5	27.8	186	Sand	20.0	9.81	10.19	290.42	282.77	0.654	0.6	1.00	1	1	186	122	68	104	133
28.5	30.0	29.3	200	Sand	20.0	9.81	10.19	305.70	298.06	0.637	0.6	1.00	1	1	200	127	71	109	139

SPT Correction																				
Project :																	Borehole No		4	
Client:																	Water Table		-0.00m	
Location:			SITE:- PLOT NO. - 4356, ROAD NO. - 23, BLOCK - N, BASHUNDHARA RESIDENTIAL AREA, DHAKA, BANGLADESH.																	
Depth in meter measured from EGL			Field SPT	Soil Type	Unit Wt. of soil (kN/m ³)	Unit Wt. of water (kN/m ³)	Effective Unit Wt. of soil (kN/m ³)	Effective overburden pressure σ_v (KN/m ²)	Avg. Effective overburden pressure σ_v (KN/m ²)	C_N	Hammer efficiency, E_H	Borehole diameter correction, C_H	Sampler correction, C_s	Rod length correction, C_R	Corrected SPT (N_{60}) after Field procedure	Corrected SPT due to Overburden Pressure (N_{60})	Correction of SPT Value for water table (Dilatancy) Value N''	N_{60}	N_{60}	
from	to	Average																		
0.0	1.5	0.8	2	Sand	20.0	9.81	10.19	15.29	7.64	1.862	0.6	1	1	0.75	2	3	3	2	3	
1.5	3.0	2.3	1	Sand	20.0	9.81	10.19	30.57	22.93	1.494	0.6	1.00	1	0.75	1	1	1	1	1	
3.0	4.5	3.8	2	Sand	20.0	9.81	10.19	45.86	38.21	1.323	0.6	1.00	1	0.85	2	2	2	2	2	
4.5	6.0	5.3	1	Sand	20.0	9.81	10.19	61.14	53.50	1.211	0.6	1.00	1	0.95	1	1	1	1	1	
6.0	7.5	6.8	2	Silt	17.0	9.81	7.19	53.93	48.53	1.244	0.6	1.00	1	0.95	2	2	2	2	3	
7.5	9.0	8.3	3	Silt	17.0	9.81	7.19	64.71	59.32	1.176	0.6	1.00	1	0.95	3	3	3	3	4	
9.0	10.5	9.8	3	Silt	17.0	9.81	7.19	75.50	70.10	1.121	0.6	1.00	1	1	3	3	3	3	4	
10.5	12.0	11.3	4	Silt	17.0	9.81	7.19	86.28	80.89	1.073	0.6	1.00	1	1	4	4	4	4	5	
12.0	13.5	12.8	4	Silt	17.0	9.81	7.19	97.07	91.67	1.031	0.6	1.00	1	1	4	4	4	4	4	
13.5	15.0	14.3	3	Silt	17.0	9.81	7.19	107.85	102.46	0.994	0.6	1.00	1	1	3	3	3	3	3	
15.0	16.5	15.8	4	Silt	17.0	9.81	7.19	118.64	113.24	0.960	0.6	1.00	1	1	4	4	4	3	4	
16.5	18.0	17.3	3	Silt	17.0	9.81	7.19	129.42	124.03	0.930	0.6	1.00	1	1	3	3	3	2	3	
18.0	19.5	18.8	38	Sand	20.0	9.81	10.19	198.71	191.06	0.785	0.6	1.00	1	1	38	30	22	26	33	
19.5	21.0	20.3	65	Sand	20.0	9.81	10.19	213.99	206.35	0.760	0.6	1.00	1	1	65	49	32	42	54	
21.0	22.5	21.8	95	Sand	20.0	9.81	10.19	229.28	221.63	0.736	0.6	1.00	1	1	95	70	42	60	76	
22.5	24.0	23.3	136	Sand	20.0	9.81	10.19	244.56	236.92	0.713	0.6	1.00	1	1	136	97	56	83	106	
24.0	25.5	24.8	139	Sand	20.0	9.81	10.19	259.85	252.20	0.692	0.6	1.00	1	1	139	96	56	83	105	
25.5	27.0	26.3	148	Sand	20.0	9.81	10.19	275.13	267.49	0.673	0.6	1.00	1	1	148	100	57	85	109	
27.0	28.5	27.8	155	Sand	20.0	9.81	10.19	290.42	282.77	0.654	0.6	1.00	1	1	155	101	58	87	111	
28.5	30.0	29.3	170	Sand	20.0	9.81	10.19	305.70	298.06	0.637	0.6	1.00	1	1	170	108	62	93	118	

SPT Correction																				
Project :																		Borehole No		3
Client:																		Water Table		-0.00m
Location:			SITE:- PLOT NO. - 4356, ROAD NO. - 23, BLOCK - N, BASHUNDHARA RESIDENTIAL AREA, DHAKA, BANGLADESH.																	
Depth in meter measured from EGL			Field SPT	Soil Type	Unit Wt. of soil (kN/m ³)	Unit Wt. of water (kN/m ³)	Effective Unit Wt. of soil (kN/m ³)	Effective overburden pressure σ_v (kN/m ²)	Avg. Effective overburden pressure σ_v (kN/m ²)	C_u	Hammer efficiency, E_H	Borehole diameter correction, C_B	Sampler correction, C_S	Rod length correction, C_R	Corrected SPT (N_{60}) after Field procedure	Corrected SPT due to Overburden Pressure (N_{60})	Correction of SPT Value for water table (Dilatancy) Value N'	N_{60}	N_{60}	
from	to	Average																		
0.0	1.5	0.8	2	Sand	20.0	9.81	10.19	15.29	7.64	1.862	0.6	1	1	0.75	2	3	3	2	3	
1.5	3.0	2.3	2	Sand	20.0	9.81	10.19	30.57	22.93	1.494	0.6	1.00	1	0.75	2	2	2	2	2	
3.0	4.5	3.8	3	Sand	20.0	9.81	10.19	45.86	38.21	1.323	0.6	1.00	1	0.85	3	3	3	3	4	
4.5	6.0	5.3	3	Sand	20.0	9.81	10.19	61.14	53.50	1.211	0.6	1.00	1	0.95	3	3	3	3	4	
6.0	7.5	6.8	1	Silt	17.0	9.81	7.19	53.93	48.53	1.244	0.6	1.00	1	0.95	1	1	1	1	1	
7.5	9.0	8.3	2	Silt	17.0	9.81	7.19	64.71	59.32	1.176	0.6	1.00	1	0.95	2	2	2	2	2	
9.0	10.5	9.8	2	Silt	17.0	9.81	7.19	75.50	70.10	1.121	0.6	1.00	1	1	2	2	2	2	2	
10.5	12.0	11.3	4	Silt	17.0	9.81	7.19	86.28	80.89	1.073	0.6	1.00	1	1	4	4	4	4	5	
12.0	13.5	12.8	3	Silt	17.0	9.81	7.19	97.07	91.67	1.031	0.6	1.00	1	1	3	3	3	3	3	
13.5	15.0	14.3	4	Silt	17.0	9.81	7.19	107.85	102.46	0.994	0.6	1.00	1	1	4	4	4	3	4	
15.0	16.5	15.8	4	Silt	17.0	9.81	7.19	118.64	113.24	0.960	0.6	1.00	1	1	4	4	4	3	4	
16.5	18.0	17.3	4	Silt	17.0	9.81	7.19	129.42	124.03	0.930	0.6	1.00	1	1	4	4	4	3	4	
18.0	19.5	18.8	6	Clay	18.0	9.81	8.19	159.71	153.56	0.858	0.6	1.00	1	1	6	5	5	4	6	
19.5	21.0	20.3	46	Sand	20.0	9.81	10.19	213.99	206.35	0.760	0.6	1.00	1	1	46	35	25	30	38	
21.0	22.5	21.8	95	Sand	20.0	9.81	10.19	229.28	221.63	0.736	0.6	1.00	1	1	95	70	42	60	76	
22.5	24.0	23.3	118	Sand	20.0	9.81	10.19	244.56	236.92	0.713	0.6	1.00	1	1	118	84	50	72	92	
24.0	25.5	24.8	130	Sand	20.0	9.81	10.19	259.85	252.20	0.692	0.6	1.00	1	1	130	90	53	77	98	
25.5	27.0	26.3	127	Sand	20.0	9.81	10.19	275.13	267.49	0.673	0.6	1.00	1	1	127	85	50	73	93	
27.0	28.5	27.8	129	Sand	20.0	9.81	10.19	290.42	282.77	0.654	0.6	1.00	1	1	129	84	50	72	92	
28.5	30.0	29.3	139	Sand	20.0	9.81	10.19	305.70	298.06	0.637	0.6	1.00	1	1	139	88	52	76	97	

SPT Correction																				
Project :																	Borehole No		2	
Client:																	Water Table		-0.00m	
Location:			SITE:- PLOT NO. - 4356, ROAD NO. - 23, BLOCK - N, BASHUNDHARA RESIDENTIAL AREA, DHAKA, BANGLADESH.																	
Depth in meter measured from EGL			Field SPT	Soil Type	Unit Wt. of soil (kN/m ³)	Unit Wt. of water (kN/m ³)	Effective Unit Wt. of soil (kN/m ³)	Effective overburden pressure σ_v (KN/m ²)	Avg. Effective overburden pressure σ_v (KN/m ²)	C_N	Hammer efficiency, E_H	Borehole diameter correction, C_H	Sampler correction, C_S	Rod length correction, C_R	Corrected SPT (N_{60}) after Field procedure	Corrected SPT due to Overburden Pressure (N_{60})	Correction of SPT Value for water table (Dilatancy) Value N_c	N_{60}	N_{60}	
from	to	Average																		
0.0	1.5	0.8	2	Sand	20.0	9.81	10.19	15.29	7.64	1.862	0.6	1	1	0.75	2	3	3	2	3	
1.5	3.0	2.3	2	Sand	20.0	9.81	10.19	30.57	22.93	1.494	0.6	1.00	1	0.75	2	2	2	2	2	
3.0	4.5	3.8	1	Sand	20.0	9.81	10.19	45.86	38.21	1.323	0.6	1.00	1	0.85	1	1	1	1	1	
4.5	6.0	5.3	3	Sand	20.0	9.81	10.19	61.14	53.50	1.211	0.6	1.00	1	0.95	3	3	3	3	4	
6.0	7.5	6.8	2	Silt	17.0	9.81	7.19	53.93	48.53	1.244	0.6	1.00	1	0.95	2	2	2	2	3	
7.5	9.0	8.3	4	Silt	17.0	9.81	7.19	64.71	59.32	1.176	0.6	1.00	1	0.95	4	4	4	4	5	
9.0	10.5	9.8	4	Silt	17.0	9.81	7.19	75.50	70.10	1.121	0.6	1.00	1	1	4	4	4	4	5	
10.5	12.0	11.3	3	Silt	17.0	9.81	7.19	86.28	80.89	1.073	0.6	1.00	1	1	3	3	3	3	4	
12.0	13.5	12.8	4	Silt	17.0	9.81	7.19	97.07	91.67	1.031	0.6	1.00	1	1	4	4	4	4	4	
13.5	15.0	14.3	4	Silt	17.0	9.81	7.19	107.85	102.46	0.994	0.6	1.00	1	1	4	4	4	3	4	
15.0	16.5	15.8	7	Silt	17.0	9.81	7.19	118.64	113.24	0.960	0.6	1.00	1	1	7	7	7	6	7	
16.5	18.0	17.3	8	Clay	18.0	9.81	8.19	147.42	141.28	0.886	0.6	1.00	1	1	8	7	7	6	8	
18.0	19.5	18.8	10	Clay	18.0	9.81	8.19	159.71	153.56	0.858	0.6	1.00	1	1	10	9	9	7	9	
19.5	21.0	20.3	62	Sand	20.0	9.81	10.19	213.99	206.35	0.760	0.6	1.00	1	1	62	47	31	40	51	
21.0	22.5	21.8	77	Sand	20.0	9.81	10.19	229.28	221.63	0.736	0.6	1.00	1	1	77	57	36	49	62	
22.5	24.0	23.3	105	Sand	20.0	9.81	10.19	244.56	236.92	0.713	0.6	1.00	1	1	105	75	45	64	82	
24.0	25.5	24.8	126	Sand	20.0	9.81	10.19	259.85	252.20	0.692	0.6	1.00	1	1	126	87	51	75	95	
25.5	27.0	26.3	122	Sand	20.0	9.81	10.19	275.13	267.49	0.673	0.6	1.00	1	1	122	82	49	70	90	
27.0	28.5	27.8	126	Sand	20.0	9.81	10.19	290.42	282.77	0.654	0.6	1.00	1	1	126	82	49	71	90	
28.5	30.0	29.3	130	Sand	20.0	9.81	10.19	305.70	298.06	0.637	0.6	1.00	1	1	130	83	49	71	90	

SPT Correction																			
Project :																	Borehole No		1
Client:																	Water Table		-0.00m
Location:			SITE:- PLOT NO. - 4356, ROAD NO. - 23, BLOCK - N, BASHUNDHARA RESIDENTIAL AREA, DHAKA, BANGLADESH.																
Depth in meter measured from EGL			Field SPT	Soil Type	Unit Wt. of soil (kN/m ³)	Unit Wt. of water (kN/m ³)	Effective Unit Wt. of soil (kN/m ³)	Effective overburden pressure σ_v (kN/m ²)	Avg. Effective overburden pressure σ_v (kN/m ²)	C_u	Hammer efficiency, E_H	Borehole diameter correction, C_b	Sampler correction, C_s	Rod length correction, C_r	Corrected SPT (N_{60}) after Field procedure	Corrected SPT due to Overburden Pressure (N_{60})	Correction of SPT Value for water table (Dilatancy) Value N_1	N_2	N_3
from	to	Avera ge																	
0.0	1.5	0.8	1	Sand	20.0	9.81	10.19	15.29	7.64	1.862	0.6	1	1	0.75	1	1	1	1	2
1.5	3.0	2.3	4	Sand	20.0	9.81	10.19	30.57	22.93	1.494	0.6	1.00	1	0.75	3	4	4	4	5
3.0	4.5	3.8	4	Sand	20.0	9.81	10.19	45.86	38.21	1.323	0.6	1.00	1	0.85	3	4	4	4	5
4.5	6.0	5.3	7	Sand	20.0	9.81	10.19	61.14	53.50	1.211	0.6	1.00	1	0.95	7	8	8	7	9
6.0	7.5	6.8	3	Silt	17.0	9.81	7.19	53.93	48.53	1.244	0.6	1.00	1	0.95	3	4	4	3	4
7.5	9.0	8.3	2	Silt	17.0	9.81	7.19	64.71	59.32	1.176	0.6	1.00	1	0.95	2	2	2	2	2
9.0	10.5	9.8	2	Silt	17.0	9.81	7.19	75.50	70.10	1.121	0.6	1.00	1	1	2	2	2	2	2
10.5	12.0	11.3	3	Silt	17.0	9.81	7.19	86.28	80.89	1.073	0.6	1.00	1	1	3	3	3	3	4
12.0	13.5	12.8	4	Silt	17.0	9.81	7.19	97.07	91.67	1.031	0.6	1.00	1	1	4	4	4	4	4
13.5	15.0	14.3	4	Silt	17.0	9.81	7.19	107.85	102.46	0.994	0.6	1.00	1	1	4	4	4	3	4
15.0	16.5	15.8	5	Clay	18.0	9.81	8.19	135.14	128.99	0.917	0.6	1.00	1	1	5	5	5	4	5
16.5	18.0	17.3	7	Clay	18.0	9.81	8.19	147.42	141.28	0.886	0.6	1.00	1	1	7	6	6	5	7
18.0	19.5	18.8	13	Clay	18.0	9.81	8.19	159.71	153.56	0.858	0.6	1.00	1	1	13	11	11	10	12
19.5	21.0	20.3	45	Sand	20.0	9.81	10.19	213.99	206.35	0.760	0.6	1.00	1	1	45	34	25	29	37
21.0	22.5	21.8	77	Sand	20.0	9.81	10.19	229.28	221.63	0.736	0.6	1.00	1	1	77	57	36	49	62
22.5	24.0	23.3	97	Sand	20.0	9.81	10.19	244.56	236.92	0.713	0.6	1.00	1	1	97	69	42	59	75
24.0	25.5	24.8	115	Sand	20.0	9.81	10.19	259.85	252.20	0.692	0.6	1.00	1	1	115	80	47	68	87
25.5	27.0	26.3	125	Sand	20.0	9.81	10.19	275.13	267.49	0.673	0.6	1.00	1	1	125	84	50	72	92
27.0	28.5	27.8	141	Sand	20.0	9.81	10.19	290.42	282.77	0.654	0.6	1.00	1	1	141	92	54	79	101
28.5	30.0	29.3	145	Sand	20.0	9.81	10.19	305.70	298.06	0.637	0.6	1.00	1	1	145	92	54	79	101

Net Allowable Bearing Capacity of Foundation using Meyerhof (Bowles) Diagram

Client :	
Project :	
Location:	SITE:- PLOT NO. - 4356, ROAD NO. - 23, BLOCK - N, BASHUNDHARA RESIDENTIAL AREA, DHAKA, BANGLADESH.
BH No	5

Soil Type	Depth (m)	N (SPT)	Corrected SPT
reference line			
Sand	1.5	2	3
Sand	3.0	2	2
Sand	4.5	2	2
Sand	6.0	2	2
Silt	7.5	3	4
Silt	9.0	1	1
Silt	10.5	1	1
Silt	12.0	2	2
Silt	13.5	4	4
Silt	15.0	4	4
Clay	16.5	5	5
Clay	18.0	7	6
Sand	19.5	40	23
Sand	21.0	46	25
Sand	22.5	94	42
Sand	24.0	155	63
Sand	25.5	165	65
Sand	27.0	175	66
Sand	28.5	186	68
Sand	30.0	200	71

Net Allowable Bearing Capacity (kPa)

$D_f \backslash B$	1.0	2.0	3.0	4.0	5.0
1.5	58.5	45.4	45.8	36.7	32.9
2.5	53.2	48.4	45.0	36.6	36.0
3.0	53.2	52.8	44.3	36.9	37.7
4.5	63.8	52.8	40.2	40.3	43.7

Footing Influence Zone (Depth, m)

$D_f \backslash B$	1.0	2.0	3.0	4.0	5.0
1.5	1<D<3.5	0.5<D<5.5	0<D<7.5	0<D<9.5	0<D<11.5
2.5	2<D<4.5	1.5<D<6.5	1<D<8.5	0.5<D<10.5	0<D<12.5
3.0	2.5<D<5	2<D<7	1.5<D<9	1<D<11	0.5<D<13
4.5	4<D<6.5	3.5<D<8.5	3<D<10.5	2.5<D<12.5	2<D<14.5

Average SPT N-Value (N')

$D_f \backslash B$	1.0	2.0	3.0	4.0	5.0
1.5	2.20	2.20	2.60	2.26	2.13
2.5	2.00	2.20	2.33	2.10	2.20
3.0	2.00	2.40	2.20	2.05	2.24
4.5	2.40	2.40	2.00	2.10	2.40

Net Allowable Bearing Capacity of Foundation using Meyerhof (Bowles) Diagram

Client :	
Project :	
Location:	SITE:- PLOT NO. - 4356, ROAD NO. - 23, BLOCK - N, BASHUNDHARA RESIDENTIAL AREA, DHAKA, BANGLADESH.
BH No	4

Soil Type	Depth (m)	N (SPT)	Corrected SPT
reference line			
Sand	1.5	2	3
Sand	3.0	1	1
Sand	4.5	2	2
Sand	6.0	1	1
Silt	7.5	2	2
Silt	9.0	3	3
Silt	10.5	3	3
Silt	12.0	4	4
Silt	13.5	4	4
Silt	15.0	3	3
Silt	16.5	4	4
Silt	18.0	3	3
Sand	19.5	38	22
Sand	21.0	65	32
Sand	22.5	95	42
Sand	24.0	136	45
Sand	25.5	139	45
Sand	27.0	148	57
Sand	28.5	155	58
Sand	30.0	170	62

Net Allowable Bearing Capacity (kPa)

D _f \ B	1.0	2.0	3.0	4.0	5.0
1.5	42.6	35.1	31.7	33.3	35.6
2.5	42.6	30.8	34.7	36.6	39.9
3.0	42.6	33.0	36.2	38.7	41.7
4.5	37.2	41.8	44.3	48.0	47.4

Footing Influence Zone (Depth, m)

D _f \ B	1.0	2.0	3.0	4.0	5.0
1.5	1<D<3.5	0.5<D<5.5	0<D<7.5	0<D<9.5	0<D<11.5
2.5	2<D<4.5	1.5<D<6.5	1<D<8.5	0.5<D<10.5	0<D<12.5
3.0	2.5<D<5	2<D<7	1.5<D<9	1<D<11	0.5<D<13
4.5	4<D<6.5	3.5<D<8.5	3<D<10.5	2.5<D<12.5	2<D<14.5

Average SPT N-Value (N')

D _f \ B	1.0	2.0	3.0	4.0	5.0
1.5	1.60	1.70	1.80	2.05	2.30
2.5	1.60	1.40	1.80	2.10	2.44
3.0	1.60	1.50	1.80	2.15	2.48
4.5	1.40	1.90	2.20	2.50	2.60

Net Allowable Bearing Capacity of Foundation using Meyerhof (Bowles) Diagram

Client :	
Project :	
Location:	SITE:- PLOT NO. – 4356, ROAD NO. – 23, BLOCK – N, BASHUNDHARA RESIDENTIAL AREA, DHAKA, BANGLADESH.
BH No	3

Soil Type	Depth (m)	N (SPT)	Corrected SPT
reference line			
Sand	1.5	2	3
Sand	3.0	2	2
Sand	4.5	3	3
Sand	6.0	3	3
Silt	7.5	1	1
Silt	9.0	2	2
Silt	10.5	2	2
Silt	12.0	4	4
Silt	13.5	3	3
Silt	15.0	4	4
Silt	16.5	4	4
Silt	18.0	4	4
Clay	19.5	6	5
Sand	21.0	46	25
Sand	22.5	95	42
Sand	24.0	118	50
Sand	25.5	130	53
Sand	27.0	127	50
Sand	28.5	129	50
Sand	30.0	139	52

Net Allowable Bearing Capacity (kPa)

D _r \ B	1.0	2.0	3.0	4.0	5.0
1.5	63.8	55.7	42.3	37.6	37.6
2.5	69.2	55.0	43.7	39.2	41.2
3.0	74.5	52.8	44.3	41.4	42.4
4.5	69.2	48.4	44.3	48.0	48.1

Footing Influence Zone (Depth, m)

D _r \ B	1.0	2.0	3.0	4.0	5.0
1.5	1<D<3.5	0.5<D<5.5	0<D<7.5	0<D<9.5	0<D<11.5
2.5	2<D<4.5	1.5<D<6.5	1<D<8.5	0.5<D<10.5	0<D<12.5
3.0	2.5<D<5	2<D<7	1.5<D<9	1<D<11	0.5<D<13
4.5	4<D<6.5	3.5<D<8.5	3<D<10.5	2.5<D<12.5	2<D<14.5

Average SPT N-Value (N')

D _r \ B	1.0	2.0	3.0	4.0	5.0
1.5	2.40	2.70	2.40	2.32	2.43
2.5	2.60	2.50	2.27	2.25	2.52
3.0	2.80	2.40	2.20	2.30	2.52
4.5	2.60	2.20	2.20	2.50	2.64

Net Allowable Bearing Capacity of Foundation using Meyerhof (Bowles) Diagram

Client :	
Project :	
Location:	SITE:- PLOT NO. - 4356, ROAD NO. - 23, BLOCK - N, BASHUNDHARA RESIDENTIAL AREA, DHAKA, BANGLADESH.
BH No	2

Soil Type	Depth (m)	N (SPT)	Corrected SPT
reference line			
Sand	1.5	2	3
Sand	3.0	2	2
Sand	4.5	1	1
Sand	6.0	3	3
Silt	7.5	2	2
Silt	9.0	4	4
Silt	10.5	4	4
Silt	12.0	3	3
Silt	13.5	4	4
Silt	15.0	4	4
Silt	16.5	7	7
Clay	18.0	8	7
Clay	19.5	10	9
Sand	21.0	62	31
Sand	22.5	77	36
Sand	24.0	105	45
Sand	25.5	126	51
Sand	27.0	122	49
Sand	28.5	126	49
Sand	30.0	130	49

Net Allowable Bearing Capacity (kPa)

$D_f \backslash B$	1.0	2.0	3.0	4.0	5.0
1.5	53.2	43.3	38.8	41.9	42.3
2.5	37.2	44.0	45.0	47.0	45.8
3.0	42.6	44.0	48.3	48.7	47.8
4.5	63.8	55.0	56.3	54.8	54.6

Footing Influence Zone (Depth, m)

$D_f \backslash B$	1.0	2.0	3.0	4.0	5.0
1.5	1<D<3.5	0.5<D<5.5	0<D<7.5	0<D<9.5	0<D<11.5
2.5	2<D<4.5	1.5<D<6.5	1<D<8.5	0.5<D<10.5	0<D<12.5
3.0	2.5<D<5	2<D<7	1.5<D<9	1<D<11	0.5<D<13
4.5	4<D<6.5	3.5<D<8.5	3<D<10.5	2.5<D<12.5	2<D<14.5

Average SPT N-Value (N')

$D_f \backslash B$	1.0	2.0	3.0	4.0	5.0
1.5	2.00	2.10	2.20	2.58	2.74
2.5	1.40	2.00	2.33	2.70	2.80
3.0	1.60	2.00	2.40	2.70	2.84
4.5	2.40	2.50	2.80	2.85	3.00

Net Allowable Bearing Capacity of Foundation using Meyerhof (Bowles) Diagram

Client :	
Project :	
Location:	SITE:- PLOT NO. - 4356, ROAD NO. - 23, BLOCK - N, BASHUNDHARA RESIDENTIAL AREA, DHAKA, BANGLADESH.
BH No	1

Soil Type	Depth (m)	N (SPT)	Corrected SPT
reference line			
Sand	1.5	1	1
Sand	3.0	4	4
Sand	4.5	4	4
Sand	6.0	7	8
Silt	7.5	3	4
Silt	9.0	2	2
Silt	10.5	2	2
Silt	12.0	3	3
Silt	13.5	4	4
Silt	15.0	4	4
Clay	16.5	5	5
Clay	18.0	7	6
Clay	19.5	13	11
Sand	21.0	45	25
Sand	22.5	77	36
Sand	24.0	97	42
Sand	25.5	115	47
Sand	27.0	125	50
Sand	28.5	141	54
Sand	30.0	145	54

Net Allowable Bearing Capacity (kPa)

D _f \ B	1.0	2.0	3.0	4.0	5.0
1.5	90.4	86.6	74.0	60.7	54.4
2.5	106.4	114.3	83.6	64.5	57.6
3.0	127.7	114.3	88.5	68.5	61.2
4.5	170.2	105.5	80.5	74.0	70.7

Footing Influence Zone (Depth, m)

D _f \ B	1.0	2.0	3.0	4.0	5.0
1.5	1<D<3.5	0.5<D<5.5	0<D<7.5	0<D<9.5	0<D<11.5
2.5	2<D<4.5	1.5<D<6.5	1<D<8.5	0.5<D<10.5	0<D<12.5
3.0	2.5<D<5	2<D<7	1.5<D<9	1<D<11	0.5<D<13
4.5	4<D<6.5	3.5<D<8.5	3<D<10.5	2.5<D<12.5	2<D<14.5

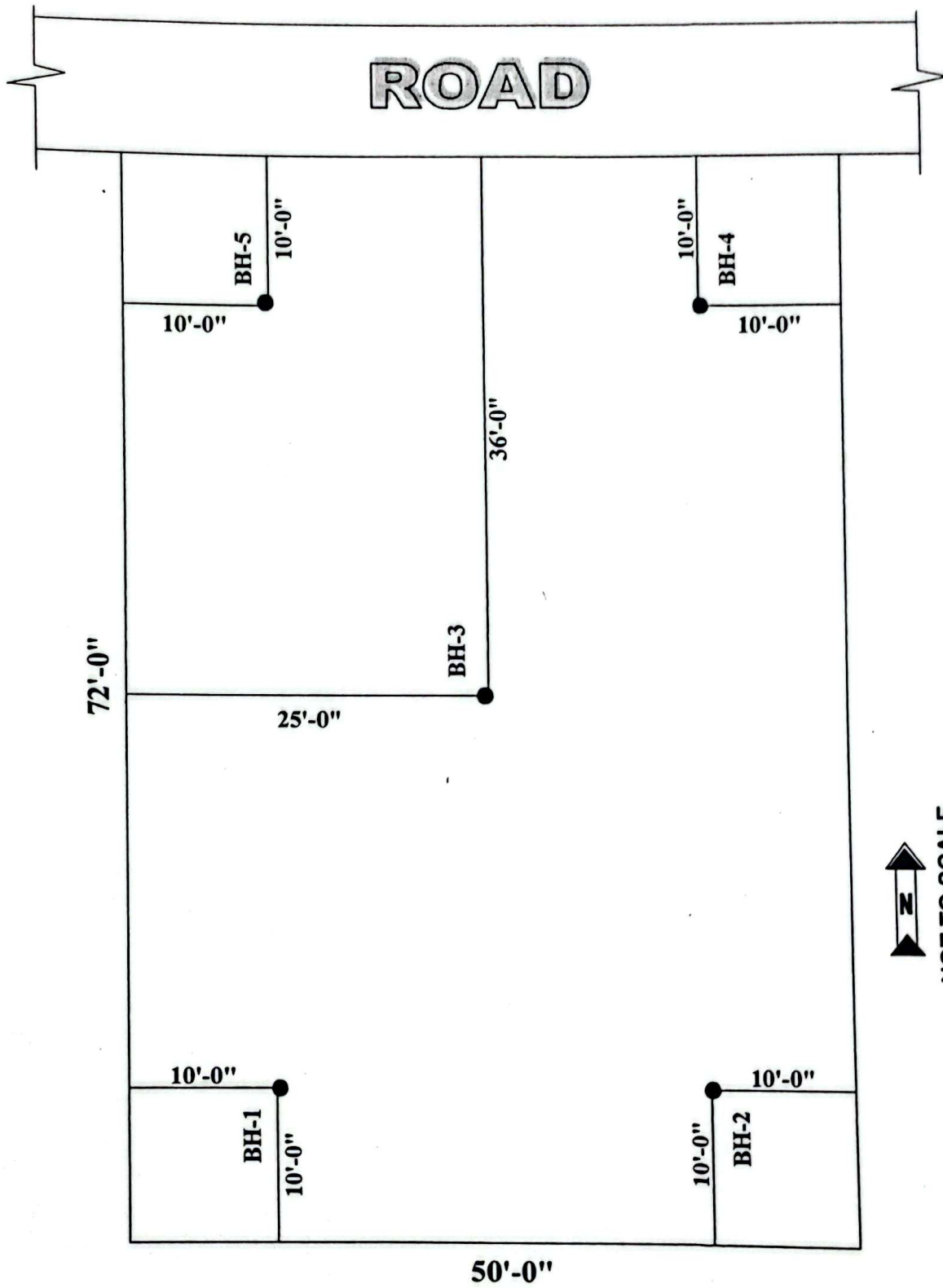
Average SPT N-Value (N')

D _f \ B	1.0	2.0	3.0	4.0	5.0
1.5	3.40	4.20	4.20	3.74	3.52
2.5	4.00	5.20	4.33	3.70	3.52
3.0	4.80	5.20	4.40	3.80	3.64
4.5	6.40	4.80	4.00	3.85	3.88

SITE:- PLOT NO. - 4356, ROAD NO. - 23, BLOCK -
N. BASHUNDHARA RESIDENTIAL AREA, DHAKA,
BANGLADESH.

SITE PLAN

MEGHLA SOIL & FOUNDATION



MEGHNA SOIL & FOUNDATION

SITE:- PLOT NO. - 4356, ROAD NO. - 23, BLOCK - N, BASHUNDHARA RESIDENTIAL AREA, DHAKA, BANGLADESH.

Date of Start : 8/8/2025

Type of Drilling : Manual drive (Hand Wash)

Date of Completion : 8/8/2025

TBM taken on top of Carpeting road, TBM = 100.00

Weather :- Sunny

RL = 0'-0"

GWL = 0'-0"

Bore Hole No. 01 (One)

Total Depth = 100' - 0"

Depth in meter	Depth in feet	Thickness in feet	Discription of Classification (Visual)	Graphics Boring Dia	S.P.T				S.P.T Graph	Grain size analysis			N.M.C. (%)	Liquid Limit (LL)	Plastic Limit (PL)	Sample Type
					15cm	15cm	15cm	30cm		Sand (%)	Silt (%)	Clay (%)				
1.5	5		Sand and Etc. Filling		1	0	1	1								D-1
3.02	10				1	2	2	4								D-2
4.54	15				1	2	2	4								D-3
6.06	20				2	2	5	7								D-4
7.58	25	22'-0"	Grey very soft to soft Clayey SILT with trace fine sand		1	1	2	3								D-5
9.1	30				1	1	1	2								D-6
10.6	35				1	1	1	2								D-7
12.1	40				1	1	2	3								D-8
13.7	45				1	2	2	4								D-9
15.2	50				1	2	2	4								D-10
16.7	55	52'-0"	Light Grey medium stiff to stiff Plastic CLAY with trace fine sand and silt		2	2	3	5								D-11
18.2	60				1	3	4	7								D-12
19.7	65				2	5	8	13								D-13
21.3	70	67'-0"			10	19	26	45								D-14
22.8	75		Light Grey dense to very dense fine SAND with trace silt		26	37	40	77								D-15
24.3	80				30	42	55	97								D-16
25.8	85				33	50	65	115								D-17
27.3	90				38	55	70	125								D-18
28.9	95				42	65	76	141								D-19
30.4	100	100'-0"			30	64	81	145								D-20

SAND SAMPLES

CLAY SAMPLES

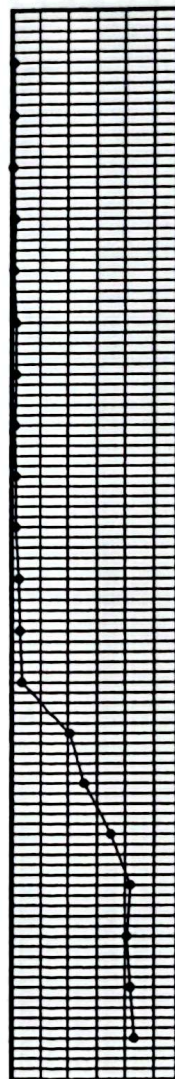
UNDISTURBED SAMPLES

MEGHNA SOIL & FOUNDATION

SITE:- PLOT NO. - 4356, ROAD NO. - 23, BLOCK - N, BASHUNDHARA RESIDENTIAL AREA, DHAKA, BANGLADESH.

Date of Start : 8/8/2025
 Date of Completion : 8/8/2025
 Weather :- Sunny
 Bore Hole No. 02 (Two)

Type of Drilling : Manual drive (Hand Wash)
 TBM taken on top of Carpeting road, TBM = 100.00
 RL = 0'-0" GWL = 0'-0"
 Total Depth = 100' - 0"

Depth in meter	Depth in feet	Thickness in feet	Discription of Classification (Visual)	Graphics	S.P.T				S.P.T Graph	Grain size analysis			N.M.C.(%)	Liquid Limit (LL)	Plastic Limit (PL)	Sample Type	
					Boring Dia	15cm	15cm	15cm		30cm	Sand(%)	Silt (%)					Clay(%)
1.5	5	23'-0"	Sand and Etc. Filling			1	1	1	2							D-1	
3.02	10					1	1	1	2							D-2	
4.54	15					1	0	1	1								D-3
6.06	20					1	1	2	3								D-4
7.58	25					1	1	1	2								D-5
9.1	30	53'-0"	Grey very soft to soft Clayey SILT with trace fine sand			1	2	2	4								D-6
10.6	35					2	2	2	4								D-7
12.1	40					1	1	2	3								D-8
13.7	45					2	2	2	4								D-9
15.2	50					2	2	2	4								D-10
16.7	55	67'-0"	Light Grey medium stiff to stiff Plastic CLAY with trace fine sand and silt			2	3	4	7								D-11
18.2	60					3	3	5	8								D-12
19.7	65					4	4	6	10								D-13
21.3	70	100'-0"	Light Grey dense to very dense fine SAND with trace silt			14	27	35	62								D-14
22.8	75					25	34	43	77								D-15
24.3	80					35	46	59	105								D-16
25.8	85					40	50	76	126								D-17
27.3	90					42	50	72	122								D-18
28.9	95					18	48	78	126								D-19
30.4	100					14	50	80	130								D-20

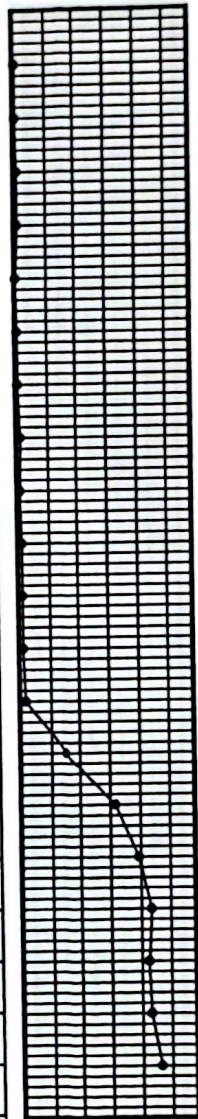
SAND SAMPLES CLAY SAMPLES UNDISTURBED SAMPLES

MEGHLA SOIL & FOUNDATION

SITE:- PLOT NO. - 4356, ROAD NO. - 23, BLOCK - N, BASHUNDHARA RESIDENTIAL AREA, DHAKA, BANGLADESH.

Date of Start : 9/8/2025
 Date of Completion : 9/8/2025
 Weather :- Sunny
 Bore Hole No. 03 (Three)

Type of Drilling : Manual drive (Hand Wash)
 TBM taken on top of Carpeting road, TBM = 100.00
 RL = 0'-0" GWL = 0'-0"
 Total Depth = 100' - 0"

Bore Hole No. 03 (Three)			Total Depth																	
Depth in meter	Depth in feet	Thickness in feet	Discription of Classification (Visual)	Graphics	S.P.T				S.P.T Graph	Grain size analysis			N.M.C. (%)	Liquid Limit (LL)	Plastic Limit (PL)	Sample Type				
					Boring Dia	15cm	15cm	15cm		30cm	Sand (%)	Silt (%)					Clay (%)			
1.5	5		Sand and Etc. Filling		1	1	1	2								D-1				
3.02	10	1			1	1	2								D-2					
4.54	15				1	1	2	3								D-3				
6.06	20				1	1	2	3								D-4				
7.58	25	23'-0"			1	0	1	1								D-5				
9.1	30		Grey very soft to soft Clayey SILT with trace fine sand		1	1	1	2									D-6			
10.6	35				1	1	1	2									D-7			
12.1	40				1	2	2	4									D-8			
13.7	45				1	1	2	3									D-9			
15.2	50				1	2	2	4									D-10			
16.7	55		Light Grey medium stiff Plastic CLAY with trace fine sand and silt		2	2	2	4									D-11			
18.2	60	61'-0"			1	2	2	4									D-12			
19.7	65				2	2	4	6									D-13			
21.3	70	67'-0"			14	20	26	46									D-14			
22.8	75				19	40	55	95									D-15			
24.3	80		Light Grey dense to very dense fine SAND with trace silt		24	50	68	118									D-16			
25.8	85				29	60	70	130									D-17			
27.3	90				33	55	72	127									D-18			
28.9	95				40	60	69	129									D-19			
30.4	100	100'-0"			42	65	74	139									D-20			

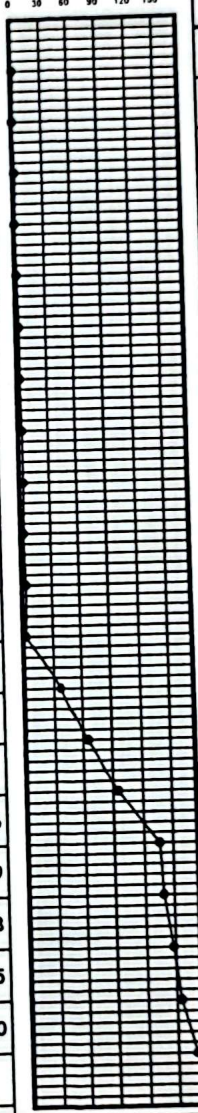
SAND SAMPLES CLAY SAMPLES UNDISTURBED SAMPLES

MEGHLA SOIL & FOUNDATION

SITE:- PLOT NO. - 4356, ROAD NO. - 23, BLOCK - N, BASHUNDHARA RESIDENTIAL AREA, DHAKA, BANGLADESH.

Date of Start : 9/8/2025
 Date of Completion : 9/8/2025
 Weather :- Sunny
 Bore Hole No. 04 (Four)

Type of Drilling : Manual drive (Hand Wash)
 TBM taken on top of Carpeting road, TBM = 100.00
 RL = 0'-0" GWL = 0'-0"
 Total Depth = 100' - 0"

Bore Hole No. 04 (Four)			Total Depth = 100'-0"														
Depth in meter	Depth in feet	Thickness in feet	Discription of Classification (Visual)	Graphics	S.P.T				S.P.T Graph	Grain size analysis			N.M.C.(%)	Liquid Limit (LL)	Plastic Limit (PL)	Sample Type	
					Boring Dia	15cm	15cm	15cm		30cm	Sand(%)	Silt (%)					Clay(%)
1.5	5	24'-0"	Sand and Etc. Filling			1	1	1	2							D-1	
3.02	10					1	0	1	1						D-2		
4.54	15					1	1	1	2							D-3	
6.06	20					1	0	1	1							D-4	
7.58	25					1	1	1	2							D-5	
9.1	30	63'-0"	Grey very soft to soft Clayey SILT with trace fine sand			1	1	2	3								D-6
10.6	35					1	1	2	3							D-7	
12.1	40					2	2	2	4							D-8	
13.7	45					1	2	2	4							D-9	
15.2	50					1	1	2	3							D-10	
16.7	55					1	2	2	4							D-11	
18.2	60					1	1	2	3							D-12	
19.7	65					10	16	22	38							D-13	
21.3	70					17	30	35	65							D-14	
22.8	75					22	40	55	95							D-15	
24.3	80	100'-0"	Light Grey dense to very dense fine SAND with trace silt			40	60	76	136								D-16
25.8	85					40	63	76	139							D-17	
27.3	90					43	69	79	148							D-18	
28.9	95					50	72	83	155							D-19	
30.4	100					50	80	90	170							D-20	
SAND SAMPLES  CLAY SAMPLES  UNDISTURBED SAMPLES 																	

SAND SAMPLES  CLAY SAMPLES  UNDISTURBED SAMPLES 

MEGHLA SOIL & FOUNDATION

SITE:- PLOT NO. - 4356, ROAD NO. - 23, BLOCK - N, BASHUNDHARA RESIDENTIAL AREA, DHAKA, BANGLADESH.

Date of Start : 10/8/2025
 Date of Completion : 10/8/2025
 Weather :- Sunny
 Bore Hole No. 05 (Five)

Type of Drilling : Manual drive (Hand Wash)
 TBM taken on top of Carpeting road, TBM = 100.00
 RL = 0'-0" GWL = 0'-0"
 Total Depth = 100' - 0"

Bore Hole No. 05 (FIVE)			Total Depth													
Depth in meter	Depth in feet	Thickness in feet	Discription of Classification (Visual)	Graphics	S.P.T				S.P.T Graph	Grain size analysis			N.M.C.(%)	Liquid Limit (LL)	Plastic Limit (PL)	Sample Type
					Boring Dia	15cm	15cm	15cm	30cm		Sand(%)	Silt (%)				
1.5	5	23'-0"	Sand and Etc. Filling		1	1	1	2								D-1
3.02	10				1	1	1	2								D-2
4.54	15				1	1	1	2								D-3
6.06	20				1	1	1	2								D-4
7.58	25				1	1	2	3								D-5
9.1	30	53'-0"	Grey very soft to soft Clayey SILT with trace fine sand		1	0	1	1							D-6	
10.6	35				1	0	1	1								D-7
12.1	40				1	1	1	2								D-8
13.7	45				1	2	2	4								D-9
15.2	50				1	2	2	4								D-10
16.7	55	63'-0"	Light Grey medium stiff Plastic CLAY with trace fine sand and silt		1	2	3	5							D-11	
18.2	60				1	3	4	7								D-12
19.7	65	100'-0"	Light Grey dense to very dense fine SAND with trace silt		11	17	23	40							D-13	
21.3	70				14	20	26	46								D-14
22.8	75				10	40	54	94								D-15
24.3	80				40	70	85	155								D-16
25.8	85				45	75	90	165								D-17
27.3	90				50	80	95	175								D-18
28.9	95				50	86	100	186								D-19
30.4	100				60	90	110	200								D-20

SAND SAMPLES CLAY SAMPLES UNDISTURBED SAMPLES

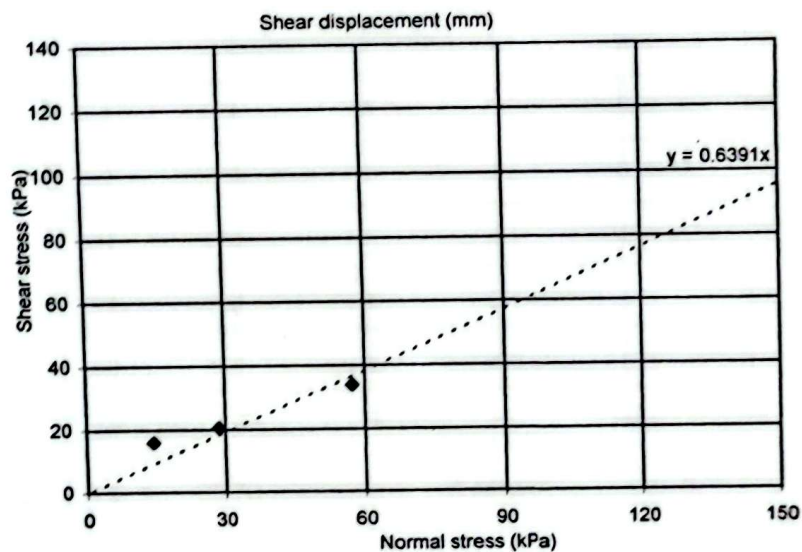
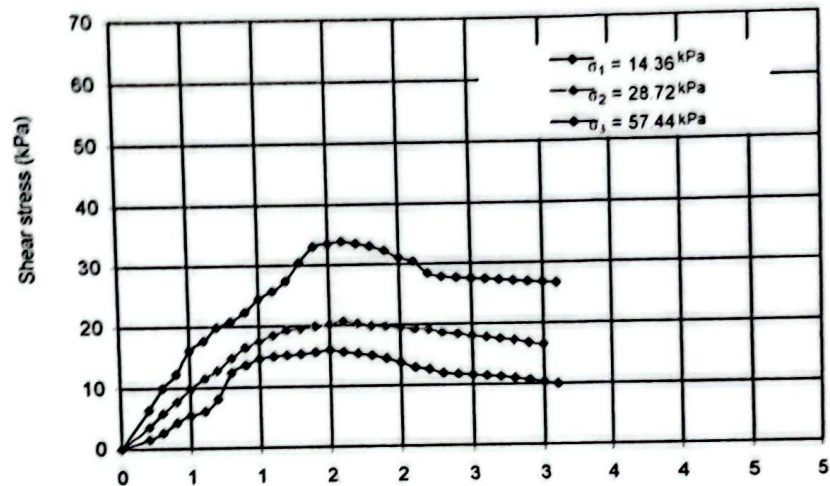
MEGHLA SOIL & FOUNDATION
DIRECT SHEAR TEST (QUICK TYPE)
Test Specification: ASTM D 2166 - 00

SITE:- PLOT NO. - 4356, ROAD NO. - 23, BLOCK - N, BASHUNDHARA RESIDENTIAL
AREA, DHAKA, BANGLADESH.

BH No. : 1

Sample ID : D-14

Depth: 70 ft



Result : $\phi = 32.58$

C (kPa) = 0

Tested By:
Engr. Md. Lakmia

Checked By:
Engr. Md. Saiful Islam

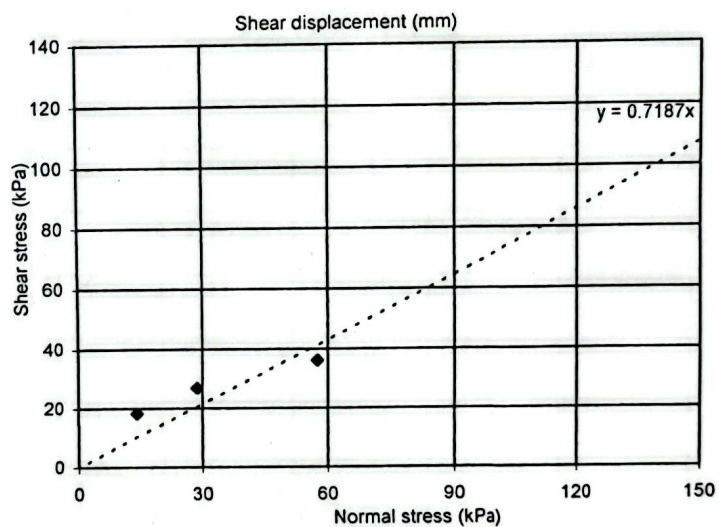
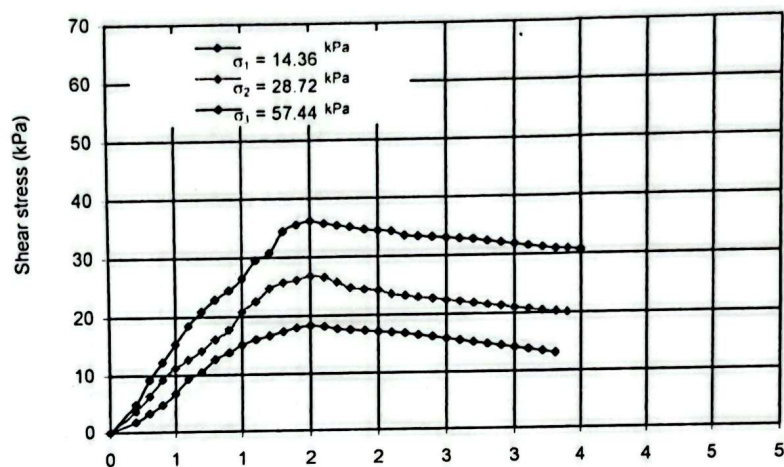
Test Specification: ASTM D 2166 - 00

SITE:- PLOT NO. - 4356, ROAD NO. - 23, BLOCK - N, BASHUNDHARA RESIDENTIAL AREA, DHAKA, BANGLADESH.

BH No. : 2

Sample ID : D-16

Depth: 80 ft



Result : $\phi = 35.70$

C (kPa) = 0

Tested By:
Engr. Md. Lakmia

Checked By:
Engr. Md. Saiful Islam

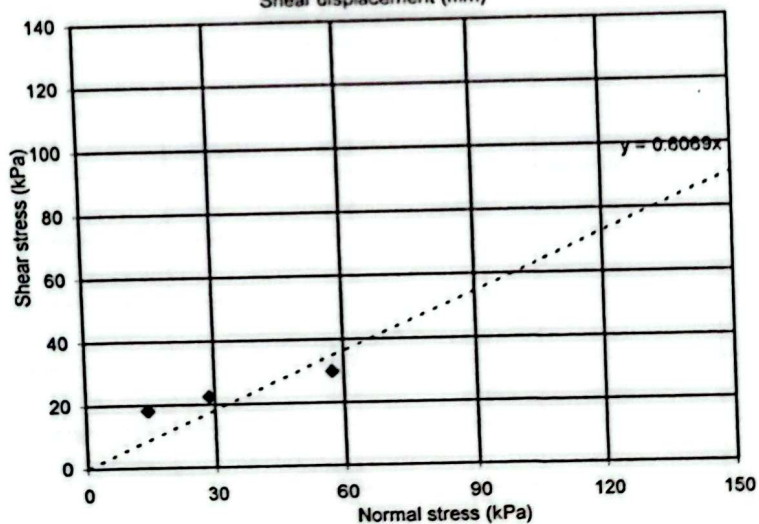
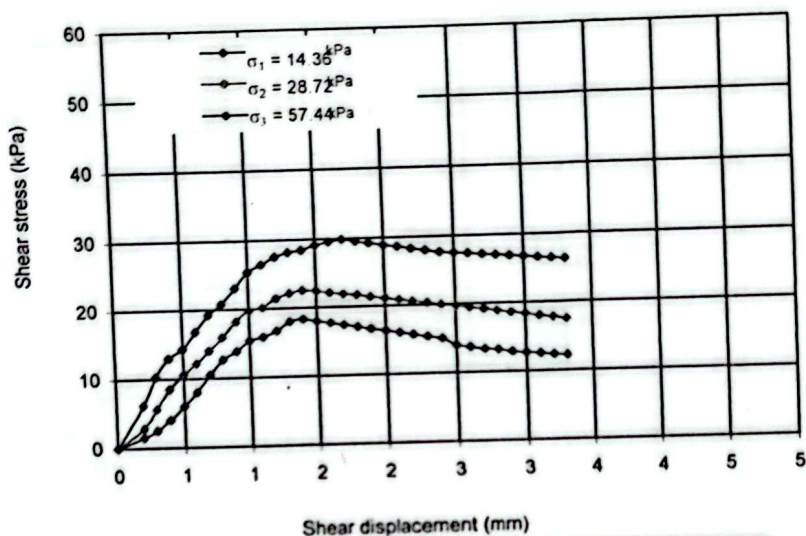
MEGHNA SOIL & FOUNDATION
DIRECT SHEAR TEST (QUICK TYPE)
Test Specification: ASTM D 2166 - 00

SITE:- PLOT NO. - 4356, ROAD NO. - 23, BLOCK - N, BASHUNDHARA RESIDENTIAL
AREA, DHAKA, BANGLADESH.

BH No. : 3

Sample ID : D-14

Depth: 70



Result : $\phi = 30.48$

$C \text{ (kPa)} = 0$

Tested By:
Engr. Md. Lakmia

Checked By:
Engr. Md. Saiful Islam

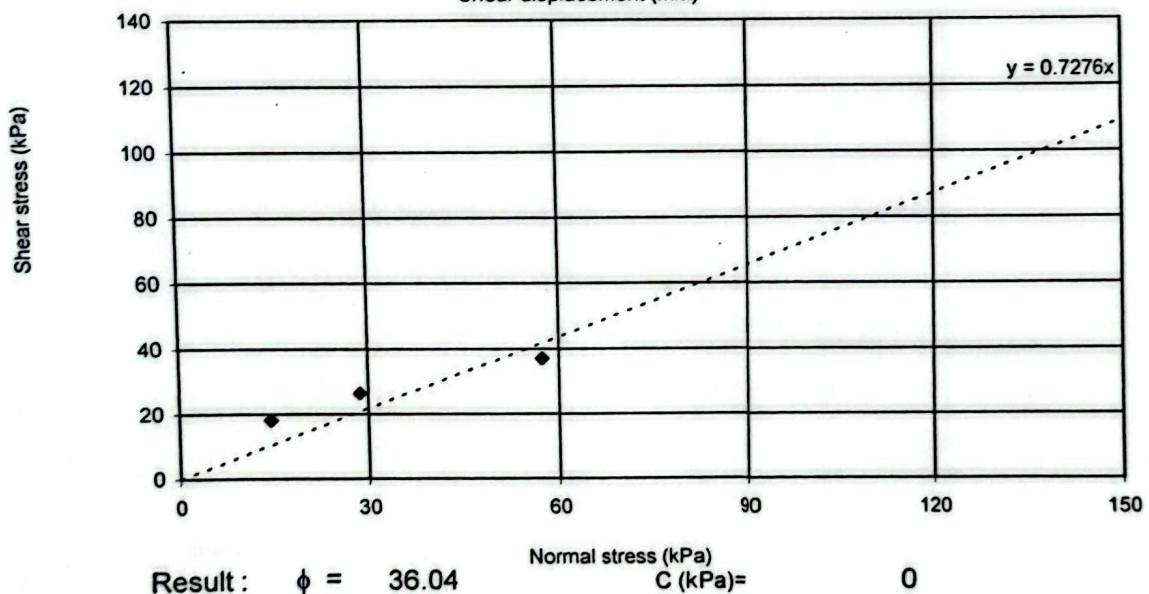
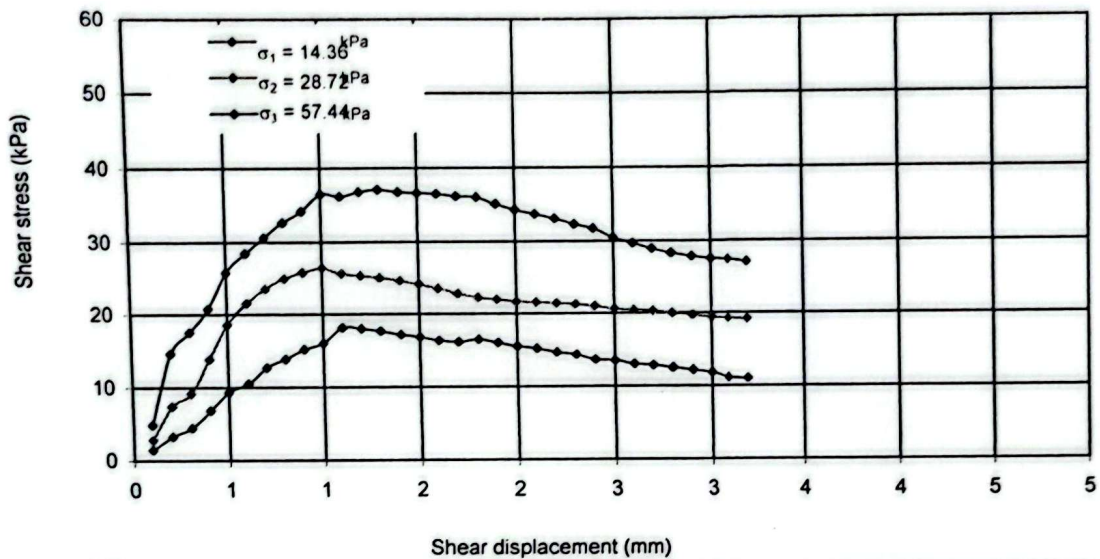
MEGHLA SOIL & FOUNDATION
DIRECT SHEAR TEST (QUICK TYPE)
Test Specification: ASTM D 2166 - 00

SITE:- PLOT NO. - 4356, ROAD NO. - 23, BLOCK - N, BASHUNDHARA RESIDENTIAL
AREA, DHAKA, BANGLADESH.

BH No. : 4

Sample ID : D-20

Depth: 100



Tested By:
Engr. Md. Lakmia

Checked By:
Engr. Md. Saiful Islam

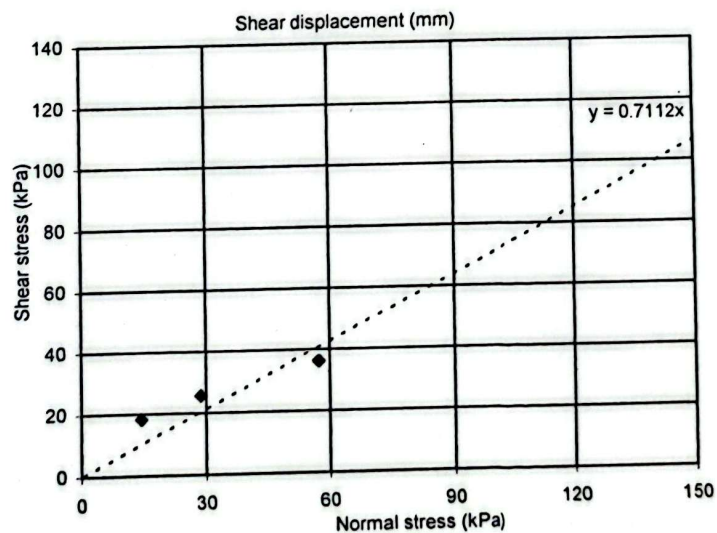
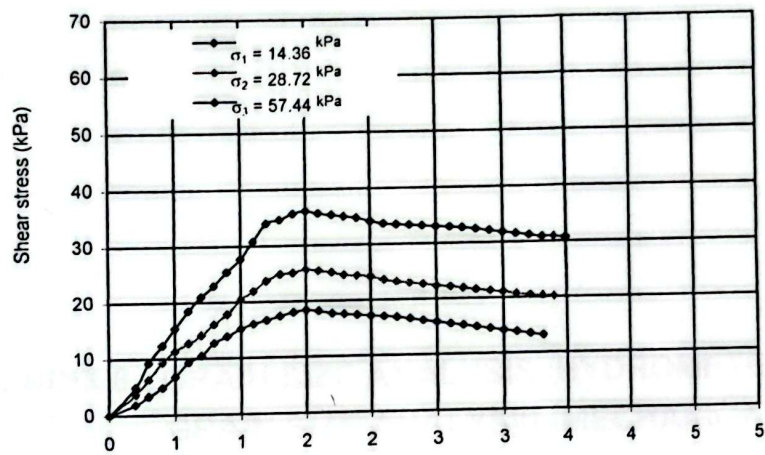
MEGHNA SOIL & FOUNDATION
DIRECT SHEAR TEST (QUICK TYPE)
Test Specification: ASTM D 2166 - 00

SITE:- PLOT NO. - 4356, ROAD NO. - 23, BLOCK - N, BASHUNDHARA RESIDENTIAL
AREA, DHAKA, BANGLADESH.

BH No. : 5

Sample ID : D-15

Depth: 75 ft



Result : $\phi = 35.42$

C (kPa) = 0

Tested By:
Engr. Md. Lakmia

Checked By:
Engr. Md. Saiful Islam